



RAPHAEL PENZ

Software Developer | Gameplay Programmer

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PROFILE

Software developer with a B.Sc. in Computer Science (Games Engineering) from the Technical University of Munich, focused on gameplay systems, software architecture and Python/C++ development. Solo developer of a published RPG/visual-novel project with 47,000+ lines of project code, custom gameplay frameworks and 2,500+ downloads. Additional experience in high-performance C++ simulation software, SIMD optimization and scalable benchmark infrastructure.

TECHNICAL PROFILE

Programming Languages	Python • C++ • Java • C#
Game Development	Ren'Py • Unity • Unreal Engine • Gameplay systems • Quest, inventory, character and combat systems
Architecture	Modular software architecture • Save/update compatibility • Data-driven development
Performance	SIMD optimization • AoS/SoA memory layouts • Scalable simulations
Tools	Git • Linux • Docker/Kubernetes (basic knowledge)
Languages	German - Native • English - C2

PROJECT EXPERIENCE

Harmony of the Stars - Solo Developer

September 2024 - present

RPG/visual novel hybrid | Python, Ren'Py | published solo project | theh0bbit.github.io

- Development and public release of a self-developed RPG/visual-novel hybrid with 2,500+ downloads.
- Implementation of core gameplay systems: combat, inventory, quests, progression systems, relationships, character planning and world navigation.
- Design of a modular gameplay architecture supporting interconnected systems, long-term maintainability and update-friendly releases.
- Implementation of a save-compatible project structure so new content and system changes can reliably support existing save files.
- Creation of 23,000+ lines of gameplay code and 47,000+ lines of total project code.
- Full responsibility for programming, system design, software architecture, worldbuilding, release management and project organization.

PROJECT EXPERIENCE - RESEARCH / C++

AutoPas - Bachelor's Thesis, Technical University of Munich

October 2021

C++ particle simulation framework | Discrete Element Method | SIMD | AoS/SoA

- Implemented a new particle type based on the Discrete Element Method (DEM) in the C++ AutoPas framework.
- Developed pairwise force calculation functors for AoS and SoA memory layouts, including SIMD optimization.
- Extended the simulation infrastructure and integrated DEM simulations into the existing benchmark suite for molecular dynamics.
- Evaluated scalability and performance using simulations with up to 1,000,000 particles.
- Compared physical behavior and runtime characteristics with existing molecular dynamics implementations.

EDUCATION

B.Sc. Computer Science: Games Engineering - TUM

Oct 2017 - Sep 2024

Technical University of Munich, Munich

M.Sc. Computer Science: Games Engineering - TUM

Oct 2024 - Sep 2025

Technical University of Munich, Munich | 87/120 ECTS completed; not completed

PROFESSIONAL EXPERIENCE

Luxury Boutique Hotel Himmlhof - Hotel Employee (part-time)

August 2021 - present

St. Anton am Arlberg, Tyrol

- Part-time role in daily hotel operations with additional IT support: setup and configuration of PCs, troubleshooting Windows and software issues, printer configuration and maintenance of the hotel website.
- Support in guest service, reception and organizational tasks; reliable communication with guests and team in a seasonal business environment.
- Flexible assumption of varied tasks according to business requirements.

Austrian Red Cross - Civilian Service / Paramedic

September 2016 - June 2017

Landeck, Tyrol

- Worked as a paramedic and supported the dispatch center with call handling, organization of patient transports and coordination of drivers and vehicles.

WORKING STYLE AND APPLICATION FOCUS

Independent implementation of complex systems • structured problem analysis • focus on maintainable architecture • interest in gameplay programming, tools/systems development and general software development